I Rosenman and C Smith. 2001. Appraisal theory: Overview, assumptions, varieties, controversies. *Appraisal processes in emotion. Theory, methods, research*, pages 3–19.

Klaus R Scherer et al. 1984. On the nature and function of emotion: A component process approach. *Approaches to emotion*, 2293(317):31.

Ashish Sharma, Kevin Rushton, Inna Lin, David Wadden, Khendra Lucas, Adam Miner, Theresa Nguyen, and Tim Althoff. 2023. Cognitive reframing of negative thoughts through human-language model interaction. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9977–10000, Toronto, Canada. Association for Computational Linguistics.

Xing Shi, Inkit Padhi, and Kevin Knight. 2016. Does string-based neural MT learn source syntax? In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1526–1534, Austin, Texas. Association for Computational Linguistics.

Beau Sievers and Mark A Thornton. 2024. Deep social neuroscience: the promise and peril of using artificial neural networks to study the social brain. *Social Cognitive and Affective Neuroscience*, 19(1):nsae014.

Craig A Smith and Phoebe C Ellsworth. 1985. Patterns of cognitive appraisal in emotion. *Journal of personality and social psychology*, 48(4):813.

Craig A Smith and Leslie D Kirby. 2011. The role of appraisal and emotion in coping and adaptation. *The handbook of stress science: Biology, psychology, and health*, pages 195–208.

Joseph Suh, Suhong Moon, Minwoo Kang, and David Chan. 2024. Rediscovering the latent dimensions of personality with large language models as trait descriptors. In *NeurIPS 2024 Workshop on Behavioral Machine Learning*.

Ala N. Tak and Jonathan Gratch. 2023. Is GPT a Computational Model of Emotion? . In *2023 11th International Conference on Affective Computing and Intelligent Interaction (ACII)*, pages 1–8, Los Alamitos, CA, USA. IEEE Computer Society.

Ala N. Tak and Jonathan Gratch. 2024. Gpt-4 emulates average-human emotional cognition from a third-person perspective.

Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, et al. 2024. Gemma 2: Improving open language models at a practical size.

Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. 2024. Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet. *Transformer Circuits Thread*.

Ian Tenney, Dipanjan Das, and Ellie Pavlick. 2019a. BERT rediscovers the classical NLP pipeline. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4593–4601, Florence, Italy. Association for Computational Linguistics.

Ian Tenney, Patrick Xia, Berlin Chen, Alex Wang, Adam Poliak, R Thomas McCoy, Najoung Kim, Benjamin Van Durme, Sam Bowman, Dipanjan Das, and Ellie Pavlick. 2019b. What do you learn from context? probing for sentence structure in contextualized word representations. In *International Conference on Learning Representations*.

Eric Todd, Millicent Li, Arnab Sen Sharma, Aaron Mueller, Byron C Wallace, and David Bau. 2024. Function vectors in large language models. In *The Twelfth International Conference on Learning Representations*.

Enrica Troiano, Laura Oberländer, and Roman Klinger. 2023. Dimensional modeling of emotions in text with appraisal theories: Corpus creation, annotation reliability, and prediction. *Computational Linguistics*, 49(1):1–72.

Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J Vazquez, Ulisse Mini, and Monte MacDiarmid. 2023. Activation addition: Steering language models without optimization. *arXiv e-prints*, pages arXiv–2308.

Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. 2020. Investigating gender bias in language models using causal mediation analysis. *Advances in neural information processing systems*, 33:12388–12401.

Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. 2023a. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations*.

Xuena Wang, Xueting Li, Zi Yin, Yue Wu, and Jia Liu. 2023b. Emotional intelligence of large language models. *Journal of Pacific Rim Psychology*, 17:18344909231213958.

Joshua Wondra and Phoebe Ellsworth. 2015. An appraisal theory of empathy and other vicarious emotional experiences. *Psychological review*, 122.

Zhengxuan Wu, Atticus Geiger, Jing Huang, Aryaman Arora, Thomas Icard, Christopher Potts, and Noah D. Goodman. 2024. A reply to makelov et al. (2023)'s "interpretability illusion" arguments.

Nutchanon Yongsatianchot, Parisa Ghanad Torshizi, and Stacy Marsella. 2023. Investigating large language models' perception of emotion using appraisal theory. In *2023 11th International Conference on Affective Computing and Intelligent Interaction Workshops and Demos (ACIIW)*, pages 1–8. IEEE.

Weixiang Zhao, Yanyan Zhao, Xin Lu, Shilong Wang, Yanpeng Tong, and Bing Qin. 2023. Is chatgpt equipped with emotional dialogue capabilities? *arXiv preprint arXiv:2304.09582*.

A Experimental Details

A.1 Dataset Details

For this project, we employed the crowd-enVENT dataset. Troiano et al. (2023) developed the dataset by asking crowdsource writers to share an event that made them feel a particular emotion. The vignettes varied in length, ranging from short sentences to longer narratives, but the language was predominantly everyday English. The participants were then asked to evaluate their subjective experiences during that event, including their perceived appraisals. Both were rated on a 5-point Likert scale with 5 representing the highest agreement.

The dataset is comprised of examples from the following list of emotions: *Joy, Pride, Surprise, Trust, Relief, Neutral, Boredom, Sadness, Fear, Guilt, Shame, Disgust, and Anger*. The dataset has 500 examples for each emotion, except for guilt and shame, which have 250 samples. The events were appraised along the following dimensions: *pleasantness, other-agency, predictability, suddenness, familiarity, unpleasantness, goal-relatedness, own responsibility, situational responsibility, goal support, consequence anticipation, urgency of response, own control, others' control, situational control, accepted control, internal standards, external norms, attention, not considered, and effort*. We selected a subset of dimensions for our analyses, previously shown to have a high association with emotions (Wondra and Ellsworth, 2015; Tak and Gratch, 2024).

To give a few examples from the dataset, the emotion label for the sentence "I baked a delicious strawberry cobbler" is *pride*, with appraisals *pleasantness* $= 5$, *other-agency* $= 1$, while "A housemate came at me with a knife" is an example of *fear* with *pleasantness* $= 1$ and *other-agency* $= 5$.

A.2 Architecture and Model Details

In this paper we experimented with ten LLMs: meta-llama/Llama-3.2-1B-Instruct and meta-llama/Llama-3.1-8B-Instruct (Grattafiori et al., 2024), google/gemma-2b-it and google/gemma-2-9b-it (Team et al., 2024), allenai/OLMo-7B-Instruct and allenai/OLMo-2-1124-13B-Instruct (OLMo et al., 2024), microsoft/Phi-3.5-mini-instruct and microsoft/Phi-3-medium-128k-instruct (Abdin et al., 2024), and mistralai/Ministral-8B-Instruct-2410 and nvidia/Mistral-NeMo-12B-Instruct (MistralAI, 2024). The architectural details for these language models are provided in Table 1. Unless stated otherwise, the default model used in this paper is Llama 3.2 1B since it is the lightest model, allowing for efficient analysis.

All models were implemented using the Hugging Face framework[2], leveraging the respective model weights, with additional integration of libraries such as TransformerLens (Nanda, 2022) to enable the hooking and intervention on hidden states and activations.

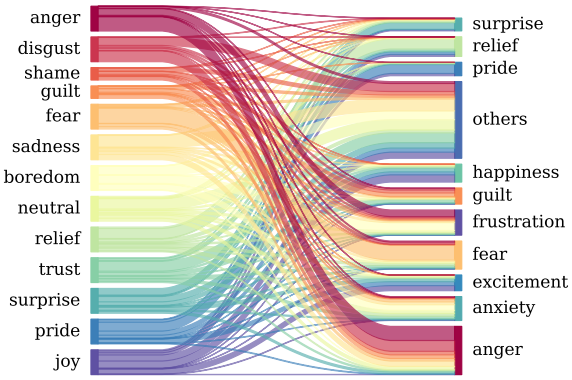

Figure 8: Llama 3.2 1B open vocab generation of emotions, comparing the true label to the predicted label through open vocab prediction

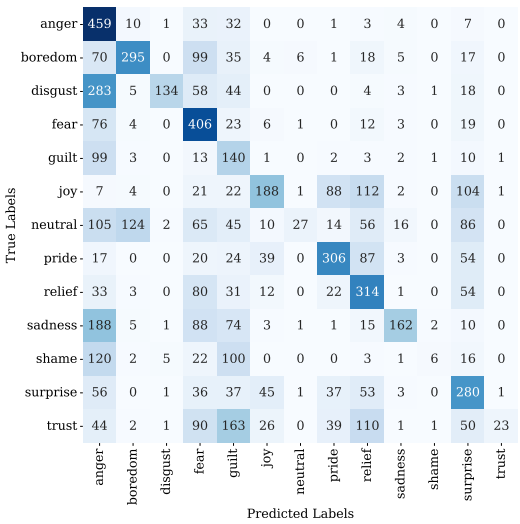

Figure 9: Confusion matrix comparing the true labels to Llama 3.2 1B predicted labels

A.3 Task Details

Emotion attribution is a challenging and subjective task, one that even humans often find difficult. The accuracy of third-person human annotations

[2] https://huggingface.co/models